

ARTIFICIAL INTELLIGENCE MEETS THE REAL WORLD

There is no end to the applications of robotics, evoking a lot of interest across different sectors of the economy. At the same time, there are fears of robots taking over humanity. It's time you get more familiar with this science in its early stages, today and in future



Laser welding by a robot

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The word 'robot' originates from the Czech word for forced labour or serf. It was introduced by playwright Karel Capek, whose fictional robotic inventions were much like Dr Frankenstein's monsters—creatures created by chemical and biological methods rather than mechanical. But the current mechanical robots of popular culture are not much different from these fictional biological creations.

The International Standards Or-

ganization (ISO) defines a robot as an automatically-controlled, re-programmable, multi-purpose and manipulative machine, with or without locomotion, for use in industrial automation applications.

Robotics is a field of multi-discipline engineering that deals with design, development and application of robots and the use of computer for their manipulation and processing.

Evolution of robotics

The history of robotics can be traced back to ancient Greece. According to Greek mythology, the Greek god of fire

and forge—Hephaestus—was served by mechanical robots. Another historical record suggests an ancient Egypt origin for robotics, where priests used steam-activated mechanisms to open the doors of their temple.

Before the first electrician was born, the earliest reference to a robot was in ancient China, in the form of an organic robot given by an artificer called Yan Shi to King Mu of Zhou around 1000 BC. The robot was made of leather and glue with actual human organs, and it ceased functioning when the organs were removed.

It was around 350 BC when veteran Greek mathematician Archytas constructed a mechanical bird named 'pigeon'—a robot powered by steam, which could fly. It was the first recorded model airplane and a milestone in the history of robotics.

Between 1500 and 1800, odd mechanical marvels were springing up all over Europe. Leonardo Da Vinci was the next known robot designer after the ancient period. He invented a mechanical man in knight's armour. John Dee of England invented a flying wooden beetle during the Elizabethan era. Another machine was Vaucanson's 'digesting duck' in 1739, which was able to annoy real ducks, quack, eat grain and produce fake faeces. The duck was even referred to by Voltaire, albeit rather cryptically: "Without the duck of Vaucanson you have nothing to remind you of the glory of France."